

Homework 4

- **Submit your solutions electronically on the course Gradescope site as PDF files.** If you plan to typeset your solutions, please use the $\text{\LaTeX}$ solution template on the course web site. If you must submit scanned handwritten solutions, please use a black pen on blank white paper and a high-quality scanner app (or an actual scanner, not just a phone camera). We will mark difficult to read solutions as incorrect and move on.
- **Every homework problem must be done *individually*.** Each problem needs to be submitted to Gradescope before midnight of the due date which can be found on the course website: <https://ecealgo.com/su26/homeworks.html>.
- For nearly every problem, **we have covered all the requisite knowledge required to complete a homework assignment prior to the “assigned” date.** This means that there is no reason not to begin a homework assignment as soon as it is assigned. Starting a problem the night before it is due is a recipe for failure.

Policies to keep in mind

- **You may use any source at your disposal**—paper, electronic, or human—but you *must* cite *every* source that you use, and you *must* write everything yourself in your own words. See the academic integrity policies on the course web site for more details.
- **Being able to clearly and concisely explain your solution is a part of the grade you will receive.** Before submitting a solution ask yourself, if you were reading the solution without having seen it before, would you be able to understand it within two minutes? If not, you need to edit. Images and flow-charts are very useful for concisely explain difficult concepts.

See the course web site (<https://ecealgo.com/su26>) for more information.

If you have any questions about these policies,
please don't hesitate to ask in class, in office hours, or on Piazza.

1. Solve the following recurrence relations. For parts (a) and (b), give an exact solution. For parts (c) and (d), give an asymptotic one. In both cases, justify your solution. Assume $A/B/C/D(1) = 1$.

(a) $A(n) = A\left(\frac{1}{2}n\right) + n^3$

(b) $B(n) = 2B\left(\frac{1}{4}n\right) + \sqrt{n}$

(c) $C(n) = C\left(\frac{1}{3}n\right) + C\left(\frac{2}{3}n\right) + O(n)$

(d) $D(n) = D\left(\frac{1}{15}n\right) + D\left(\frac{1}{10}n\right) + 2D\left(\frac{1}{6}n\right) + \sqrt{n}$

2. **The Librarian's Curse: Entropy Drift and the Phantom Log**

After the Great Fire of Dewey, the Grand Archivist recovered a list of n enchanted books, each with:

- A unique positive integer ID $S[i]$ indicating its identity, and
- A float $T[i]$ representing its ambient shelf temperature at the time of scanning.

Books were hastily placed on a linear temporal shelf. Unfortunately, due to the influence of entropic shelving fields, the order is flawed. Every time a book with a heavier ID appears *before* a lighter one ($i < j$ but $S[i] > S[j]$), the system accumulates one unit of **entropy drift**.

The total **Entropy Drift Score (EDS)** is the number of such misorderings.

Your task: Compute the Entropy Drift Score for a given shelf configuration S .

Example:

$$S = [4, 1, 3, 2], \quad T = [98.1, 97.2, 99.5, 96.0]$$

$$\text{EDS} = 4$$

Explanation: The discordant (misordered) index pairs are (0,1), (0,2), (0,3), and (2,3).

Note: An incomplete formula was recovered from the ruins:

$$\boxed{EDS = \sum_{i < j} \mathbb{1}[S[i]S[j]] + \log(T[i] + 1)}$$

Unfortunately, the log term has been deemed meaningless — you may discard it.

Warning: Any brute-force solution will awaken the Archivist's segmentation fault. Use recursive wisdom instead.